

Read quality and full genome alignment after KRAS suppression

WEEK 2 PROGRESS REPORT

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Background and progress

RNA sequencing can reveal responses to KRAS small interfering RNA (siRNA) in pancreatic ductal adenocarcinoma. All 17 full-run alignments are complete; KRAS M1 requires investigation before quantification.

Objective

Assess read quality and paired-alignment suitability before gene quantification.

Methods

FastQC 0.12.1 assessed 34 files; MultiQC 1.35 combined 32 Galaxy reports and two verified local recovery reports. HISAT2 2.2.3 aligned paired reads to human hg38 using Phred+33, unstranded mode, no input limit, no end trimming and default alignment options. RSeQC strand inference supported unstranded libraries for Control M1 and KRAS P1 only; this setting was applied to the remaining runs without individual confirmation.

Full alignment results for 17 runs

Sample	Input pairs	Overall (%)	Unique (%)	Discordant (%)
Control M1	40,237,116	96.75	86.11	0.61
Control M2	45,764,716	96.78	85.19	0.58
Control M3	38,181,985	96.67	87.14	0.64
Control P1	45,946,458	96.73	87.25	0.57
Control P2	39,989,452	95.79	86.47	1.15
Control P3	47,840,900	96.41	87.03	0.54
Control P4	46,287,075	96.20	87.22	0.65
Control P5	45,023,981	96.75	88.32	0.57
KRAS M1*	45,292,678	96.51	0.0030	75.56
KRAS M2	44,032,049	96.74	85.25	0.62
KRAS M3	42,961,283	96.64	86.95	0.60
KRAS P1	39,963,282	96.52	87.11	0.62
KRAS P2	42,632,214	96.49	87.50	0.69
KRAS P3	42,073,244	96.57	86.11	0.56
KRAS P4	40,452,536	96.77	88.32	0.52
KRAS P5	37,886,040	96.53	87.55	0.64
KRAS P5_v2	43,166,706	97.72	88.09	0.32

Table notes

Overall uses individual reads; unique and discordant use input pairs. Unique means concordantly aligned once.
 *KRAS M1 is flagged. P5_v2 is SRR24828472; KRAS P5 is SRR24828471. Total: 727,731,715 pairs. Million-pair pilots are excluded.

Read quality and alignment interpretation

All files have 150-base reads and pass base-quality and adapter checks. Composition and duplication flags occur in all 34 files; duplication is 57.0%–67.3%. Both SRR24828471 mates have GC-distribution warnings. These flags do not alone establish contamination or justify removing duplicate RNA reads. Full-run overall alignment ranges from 95.79% to 97.72%. KRAS M1 nevertheless has only 1,371 uniquely concordant pairs and 75.56% discordant pairs. Its 96.51% overall rate includes discordant and separately mapped reads and does not establish correct pairing.

Limitations and next steps

Check KRAS M1 mate identities, read order and paired-end settings before counting genes. The cause remains unresolved; retain its outputs but hold this run from downstream analysis. Compare treatments within the eight cell lines, not across unmatched primary and metastatic groups. Resolve the extra MiaPaca-2 run's repeat type before merging or treating it as independent. No differential-expression or treatment-effect conclusions are available.

References

1. Galaxy summaries for collections 253 and 156, datasets 193/195, MultiQC 165 and RSeQC outputs; 22 September 2026. Data and checks: github.com/yichuanlin07/pdac-kras-rnaseq.
2. HISAT2 manual. daehwankimlab.github.io/hisat2/manual/
3. FastQC documentation. bioinformatics.babraham.ac.uk/projects/fastqc/
4. RSeQC documentation. rseqc.sourceforge.net/
5. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.

Author contributions

Yichuan Lin organized the data and coordinated the analysis and report preparation.

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